

Dynamic Programming ^{*}

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^{*}For more discussions, please refer to the textbooks and the lecture notes of Professor Zhang. Comments are welcome.

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1 Sequential Problem

Our **sequential problem**(SP) is

$$\begin{aligned} \max_{\{u_t\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t r(x_t, u_t) \\ \text{s.t.} \quad & x_{t+1} = g(x_t, u_t) \\ & \text{with } x_0 \text{ given.} \end{aligned} \tag{1}$$

An Example:

$$\begin{aligned} \max_{\{c_t\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t \log(c_t) \\ \text{s.t.} \quad & k_{t+1} = Ak_t^\alpha - c_t \\ & \text{with } k_0 \text{ given.} \end{aligned} \tag{2}$$

2 Basic Ideas

Instead of solving the problem at once, we can figure it out recursively, period by period. It is quite similar to solve the following problem with two steps.

$$\max_{x_1, x_2} f(x_1, x_2).$$

- For problem $\max_{x_2} f(x_1, x_2)$ for any fixed x_1 , we can get $x_2^*(x_1)$ as a function of x_1 .
- Solving the problem $\max_{x_1} f(x_1, x_2^*(x_1))$, we get the optimal x_1^* , and $x_2^* = x_2^*(x_1^*)$.

Formally, a dynamic programming problem is a problem with the following structure, such that one can capture all relevant information for the decision maker in the state variables.

1. **State space** $\mathbb{X}$. $x \in \mathbb{X}$ is called a state variable.
 - (a) x_t must be relevant to the optimization problem at t .
 - (b) Once the level of the state variable is known, it does not matter how it is obtained in the past.
 - (c) If $x_t = x_s$ for $t \neq s$, then the problem starting at t is the same as starting at s .
2. A control variable u .
3. A transition law $x_{t+1} = g(x_t, u_t)$.
4. Period return $F(x_t, x_{t+1}) \equiv r(x_t, u_t)$, given the transition law.

Since the original problem can be solved recursively, we are now seeking a time-invariant policy function $h(\cdot)$ mapping the state x_t into the control u_t , such that $\{u_t\}_{t=0}^{\infty}$ generated by iterating the two functions

$$\begin{cases} u_t = h(x_t) \\ x_{t+1} = g(x_t, u_t) \end{cases}$$

starting from initial condition x_0 at $t = 0$ solves the original problem.

3 Functional Equation

The definition of value function is given by

$$\begin{aligned} V(x_0) = \max_{\{u_t\}_{t=0}^{\infty}} & \sum_{t=0}^{\infty} \beta^t r(x_t, u_t) \\ \text{s.t.} & \quad x_{t+1} = g(x_t, u_t) \\ & \quad \text{with } x_0 \text{ given.} \end{aligned} \tag{3}$$

HOW CAN WE SOLVE THE OPTIMAL PROBLEM EMPLOYING THE TOOL OF VALUE FUNCTION?

- Suppose now we knew the form of the value function $V(x)$, we can solve $h(\cdot)$ by solving for each $x \in \mathbb{X}$, the problem

$$\begin{aligned} \max_u & \{r(x, u) + \beta V(\tilde{x})\} \\ \text{s.t.} & \quad \tilde{x} = g(x, u) \\ & \quad \text{with } x \text{ given.} \end{aligned} \tag{4}$$

- But we do not know $V(\cdot)$. So we need to solve for $V(\cdot), h(\cdot)$ jointly, which are linked by Bellman equation:

$$V(x) = \max_u \{r(x, u) + \beta V[g(x, u)]\}. \tag{5}$$

Equation (5) is a **functional equation (FE)** to be solved for the pair of unknown functions $V(\cdot), h(\cdot)$.

An Example: The functional equation for Problem (2) is

$$V(k) = \max_c \{\log(c) + \beta V(Ak^\alpha - c)\}.$$

4 Formal Theorems

We begin our discussion with establishing some notations.

- $\Gamma(x)$: the set of feasible values for the state variable next period if the current state is x .

- $\Pi(x_0)$: the set of all sequences $\{x_t\}$ satisfying the constraints in **(SP)**.
Let $\vec{x} = (x_0, x_1, \dots)$ denote a typical element of $\Pi(x_0)$.
- For each $n = 0, 1, \dots$, define $u_n : \Pi(x_0) \rightarrow \mathbb{R}$ by

$$u_n(\vec{x}) = \sum_{t=0}^n \beta^t F(x_t, x_{t+1}).$$

- Define $u : \Pi(x_0) \rightarrow \bar{\mathbb{R}}$ by

$$u(\vec{x}) = \lim_{n \rightarrow \infty} u_n(\vec{x})$$

- Define the supremum function $v^* : \mathbb{X} \rightarrow \bar{\mathbb{R}}$ by

$$v^*(x_0) = \sup_{\vec{x} \in \Pi(x_0)} u(\vec{x})$$

ASSUMPTION 1 $\Gamma(x)$ is nonempty, for all $x \in \mathbb{X}$.

ASSUMPTION 2 For all $x_0 \in \mathbb{X}$ and $\vec{x} \in \Pi(x_0)$, $\lim_{n \rightarrow \infty} \sum_{t=0}^n \beta^t F(x_t, x_{t+1})$ exists.

DEFINITION 1 (v^* SATISFIES **(FE)**) We say that v^* satisfies **(FE)** if three conditions hold:

1. If $|v^*(x_0)| < \infty$, then

$$v^*(x_0) \geq F(x_0, y) + \beta v^*(y), \text{ all } y \in \Gamma(x_0)$$

and for any $\varepsilon > 0$,

$$v^*(x_0) \leq F(x_0, y) + \beta v^*(y) + \varepsilon, \text{ some } y \in \Gamma(x_0)$$

2. if $v^*(x_0) = +\infty$, then there exists a sequence y^k in $\Gamma(x_0)$ such that

$$\lim_{k \rightarrow \infty} F(x_0, y^k) + \beta v^*(y^k) = +\infty$$

3. if $v^*(x_0) = -\infty$, then

$$F(x_0, y) + \beta v^*(y) = -\infty, \text{ all } y \in \Gamma(x_0).$$

LEMMA 1 Let $\mathbb{X}, \Gamma, F$ and β satisfy Assumption 2. Then for any $x_0 \in \mathbb{X}$ and any $(x_0, x_1, \dots) = \vec{x} \in \Pi(x_0)$,

$$u(\vec{x}) = F(x_0, x_1) + \beta u(\vec{x}')$$

where $\vec{x}' = (x_1, x_2, \dots)$.

THEOREM 2 (THEOREM 4.2 IN SLP (STOKEY, LUCAS AND PRESCOTT))
Let $\mathbb{X}, \Gamma, F$ and β satisfy Assumptions 1-2. Then v^* satisfies **(FE)**.

THEOREM 3 (THEOREM 4.3 IN SLP) *Let $\mathbb{X}, \Gamma, F$ and β satisfy Assumptions 1-2. If v^* satisfies **(FE)** and satisfies*

$$\lim_{n \rightarrow \infty} \beta^n v(x_n) = 0 \text{ all } (x_0, x_1, \dots) \in \Pi(x_0), \text{ all } x_0 \in \mathbb{X},$$

then $v = v^$.*

Notes: There are two other theorems characterizing optimal feasible plans, take a look at Theorem 4.4 and Theorem 4.5 in SLP.

5 Solution Methods

We have three computational methods to solve the functional equation.

Value Function Iteration

We construct the value functions sequence by iterating on the following equation, starting from $V_0 = 0^1$, and continuing until convergence:

$$V_{j+1}(x) = \max_u \{r(x, u) + \beta V_j(\tilde{x})\}$$

subject to $\tilde{x} = g(x, u)$, x given.

DEFINITION 2 (CONTRACTION MAPPING OPERATOR) Operator T is defined by

$$(Tf)(x) = \max_{y \in \Gamma(x)} \{F(x, y) + \beta f(y)\}$$

THEOREM 4 (VALUE FUNCTION ITERATION) *If the per-period return function is bounded, then T maps from a bounded function to a bounded function. $T^j(v)$ converges to a fixed point of T uniformly for any bounded function v . This fixedpoint is bounded and hence it is V^* .*

An Example:

$$\begin{aligned} V(k_t) &= \max_{c_t} \{\ln(c_t) + \beta V(k_{t+1})\} \\ \text{s.t. } k_{t+1} &= Ak_t^\alpha - c_t; k_0 \text{ given.} \end{aligned}$$

Solution:

- Suppose $V_0(k) = 0$, we find $V_1(k)$ by solving

$$\begin{aligned} V_1(k) &= \max_c \{\ln(c) + \beta V_0(\tilde{k})\} \\ \text{s.t. } \tilde{k} &= Ak^\alpha - c; k \text{ given} \\ V_0(\tilde{k}) &= 0. \end{aligned}$$

We can get

$$V_1(k) = \ln(A) + \alpha \ln(k).$$

¹It is not necessary to be 0. We use 0 here for simplicity.

- We can now get V_2 by solving

$$\begin{aligned} V_2(k) &= \max_c \{ \ln(c) + \beta V_1(\tilde{k}) \} \\ s.t. \tilde{k} &= Ak^\alpha - c; k \text{ given} \\ V_1(\tilde{k}) &= \ln(A) + \alpha \ln(\tilde{k}). \end{aligned}$$

And we have

$$V_2(k) = \ln\left(\frac{Ak^\alpha}{1 + \alpha\beta}\right) + \beta \ln\left(Ak^\alpha - \frac{Ak^\alpha}{1 + \alpha\beta}\right)$$

- Similarly, we can get $V_3, \dots$

Finally, we can get the solution as

$$V(k) = \frac{1}{1 - \beta} \left(\frac{1}{1 - \alpha\beta} \ln(A) + \ln(1 - \alpha\beta) + \frac{\alpha\beta}{1 - \alpha\beta} \ln(\alpha\beta) \right) + \frac{\alpha}{1 - \alpha\beta} \ln(k)$$

Guess and Verify

You can guess a form of V and then verify your guessing. This method relies on the uniqueness of the solution to the equation.

An Example:

$$\begin{aligned} V(k_t) &= \max_{c_t} \{ \ln(c_t) + \beta V(k_{t+1}) \} \\ s.t. k_{t+1} &= Ak_t^\alpha - c_t; k_0 \text{ given.} \end{aligned}$$

Solution:

Let us guess the solution is

$$V(k) = E + F \ln(k)$$

Then we need to solve the following problem:

$$\begin{aligned} E + F \ln(k) &= \max_c \{ \ln(c) + \beta [E + F \ln(\tilde{k})] \} \\ s.t. \tilde{k} &= Ak^\alpha - c, \text{ with } k \text{ given.} \end{aligned}$$

The F.O.C for this problem is

$$\frac{1}{c} + \frac{\beta F}{Ak^\alpha - c}(-1) = 0$$

So we get

$$\begin{cases} c = \frac{Ak^\alpha}{1 + \beta F} \\ \tilde{k} = \frac{\beta F}{1 + \beta F} Ak^\alpha \end{cases}$$

Substitute this result back into the value function we have:

$$E + F \ln(k) = \ln \left(\frac{Ak^\alpha}{1 + \beta F} \right) + \beta \left[E + F \ln \left(\frac{\beta F}{1 + \beta F} Ak^\alpha \right) \right]$$

So we can get

$$\begin{cases} E = \frac{1}{1-\beta} \left(\frac{1}{1-\alpha\beta} \ln(A) + \ln(1 - \alpha\beta) + \frac{\alpha\beta}{1-\alpha\beta} \ln(\alpha\beta) \right) \\ F = \frac{\alpha}{1-\alpha\beta} \end{cases}$$

And also

$$\begin{cases} c_t = (1 - \alpha\beta) Ak_t^\alpha \\ k_{t+1} = \alpha\beta Ak_t^\alpha \end{cases}$$

Howard's Improvement Algorithm

Howard's improvement algorithm consists of the following steps:

1. Pick a feasible policy, $u = h_0(x)$ and compute the associated value

$$V_{h_j}(x) = \sum_{t=0}^{\infty} \beta^t r[x_t, h_j(x_t)]$$

where $x_{t+1} = g[x_t, h_j(x_t)]$, with $j = 0$.

2. Generate a new policy $u = h_{j+1}(x)$ that solves the problem

$$\max_u \{r(x, u) + \beta V_{h_j}[g(x, u)]\}$$

for each x .

3. Iterate over j to convergence.

An Example:

$$\begin{aligned} V(k_t) &= \max_{c_t} \{\ln(c_t) + \beta V(k_{t+1})\} \\ \text{s.t. } k_{t+1} &= Ak_t^\alpha - c_t; \quad k_0 \text{ given.} \end{aligned}$$

Solution:

- Let us suppose a policy form $k_{t+1} = \gamma Ak_t^\alpha$, then we should have

$$J_0(k_0) = \sum_{t=0}^{\infty} \beta^t \ln(Ak^\alpha - \gamma Ak_t^\alpha) = C + \frac{\alpha}{1 - \alpha\beta} \ln(k_0)$$

- Get the new policy $k' = \alpha\beta Ak^\alpha$, by solving

$$\max_{k'} \ln(Ak^\alpha - k') + \beta J_0(k')$$

- You can check that $k_{t+1} = \alpha\beta Ak_t^\alpha$ has already converged, so we get the solution.

6 Euler Equation and Transversality Conditions

7 Applications and Examples

References

- [1] Lars Ljungqvist, and Thomas J.Sargent, 2004, *Recursive Macroeconomic Theory*, 2nd Edition, The MIT Press.
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- [3] Andreu Mas-Collel, Michael Whinston, and Jerry R. Green, 1995, *Microeconomic Theory*, Oxford University Press.